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Chapter 20
Drawing Out; Non-Static Multi-Street Games

In the previous chapter, we discussed some clairvoyant multi-street games where the hand values do not change from street to street. We saw that in that case the non-clairvoyant player checked and called on each street as though it were a single street game. The clairvoyant player employed a strategy of "stacking" bluffs, and giving up on some of them on successive streets, such that on the river he had an appropriate mixture of hands for the single-street river strategy.

In this chapter, we continue to add more complexity by adding draws to the games. What this really means is that one or the other player may have a hand which has some probability of being best after the next street. In some cases both players will know what cards have come (we call this an ***open draw).*** This is more like a situation in holdem, where the river card is faceup and shared. We also have ***closed draws,*** which are similar to seven-card stud, where the river cards are revealed to only one player.

We begirt by considering once again a clairvoyant game. This game actually contains no draws; instead, it might be the end position of a hand where there were draws,

Example 20.1 - The Clairvoyance Game (Open Draws)
Consider the following game:

It's limit holdem and the pot contains 6 big bets.
X has an exposed pair of aces.
Y has a hand from a distribution that contains 7/8 hands
with two hearts and 1 /8 dead hands.

The board has come:
Q♥ 7♦ K♥ 4c 8♥

Using the solutions from the original clairvoyance game (Example 11.1), we can see that Y will value bet all of his flushes, and then bluff enough dead hands so that X is indifferent to calling. Using the familiar ratios we calculated when studying that game, we have the following strategies:

X checks and Y bets all of his flushes ($^7/_8$) and a times as many dead hands, or $^1/_8$ of his hands total. X folds α of his hands ($^1/_7$***)*** and calls with the remaining 6/7.

Example 20.2
But now consider this related game:

It's the turn in limit holdem and the pot contains 4 big bets.
X has an exposed pair of aces.
Y has a hand from a distribution that contains
$^1/_{10}$ hands with two hearts and $^9/_{10}$ dead hands.

For simplicity's sake, suppose the flush draw comes in exactly 20% of the time. We'll also ignore the fact that the flush comes in less frequently when Y actually has the Bush draw because there are fewer hearts left in the deck.

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The board has come:
Q♥ 7♦ K♥ 4♣

How will the action on the turn go? In Part II. we looked at games like this. So it should be clear that the aces will bet. Since Y has no hands that are ahead of the aces, he will simply call when he has pot odds (with his flush draws). He will also call with some dead hands, hoping to bluff if the flush comes in. The total number of dead hands he calls with is equal to the amount of hands he will bluff on the river if the flush comes in. We can find the number of bluffs by figuring out what the pot size will be. After a bet and a call on the turn, the pot contains six big bets. Y will want to bluff on the river (if the flush card comes) with $\alpha = 1/(P+1) = 1/7$ as many hands as he will value bet, or $1/70$ of his hands total.

Now the river comes the 8♥. We have now arrived at the situation in Example 20.1 above: the pot contains six bets, Y's distribution contains a mixture of flushes and dead hands, and the board is identical. Y now cheerfully bets all his flushes and bluffs with $1/7$ as many dead hands - which is all the dead hands he brought to the river. Before going on, you should try to figure out how often X has to call here. The single-street solution is of course that X should call $6/7$ of the time. However, the fact that we paused and turned it into a quiz should alert you that this is probably not the right answer. Suppose X decided not to call at all. Y did, after all, bet an amount on the river that would make X indifferent to calling or folding.

We know, however, that if X did decide to simply fold all the time, Y could likely exploit that. Since Y is left with only the set of hands that he wants to have to play optimally on the river, he would have to change his play on a previous street to carry forward more bluffs. But now he has a problem. The problem is that in order to carry forward bluffs to the river, Y has to call a bet on the turn with a dead hand in order to bluff on the end. However, $4/5$ of the time, he will be unable to bluff because the flush doesn't actually make it and he has no credible threat to have a strong hand. On those hands, he just loses one bet. It turns out that he can in fact do this if X never calls on the river when the flush comes in. He loses 1 bet $4/5$ of the time, but gains 5 bets when the flush comes in, for a total win of $1/5$ of a bet.

So X must find a different strategy. What he is trying to do here is not to make the flush draws indifferent on the river to bluffing with a dead hand, but to make the dead hands indifferent to the entire play of calling the turn (with nothing), hitting the flush, and bluffing the river. We can find this value algebraically. Let x be X's calling percentage. If the dead hands fold immediately on the turn, their expectation is 0. To make them indifferent to the entire play, the expectation of that play must also be zero. The dead hands lose 2 bets when the flush hits and they are called, win 5 bets (4 in the pot, plus the bet from the opponent on the turn) when they are not called, and lose 1 bet when the flush misses.

$\langle$dead hand, fold$\rangle = 0$
$\langle$dead hand, play$\rangle = p(\text{call})p(\text{flush})(-2) + p(\text{flush})p(\text{fold})(5) + p(\text{no flush})(-1)$
$\langle$dead hand, play$\rangle = x(1/5)(-2) + (1/5)(1 - x)(5) + 4/5(-1)$
$\langle$dead hand, play$\rangle = x(-7/5) + 1/5$
$x(-7/5) + 1/5 = 0$
$x = 1/7$

This leads to two principles that are at the core of analyzing multi-street play.

The first is:

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In order for a bluff to be credible, it must come from a distribution of hands that contains strong hands as well.

When considering calling ratios, the cost of playing the distribution to that point must be considered. Marked made hands can call less often on the river against marked draws when the draw comes in if the marked made hands bet and were called on previous streets. The made hands effectively extracted their value on an earlier street, and the opponent cannot exploit the made hand effectively because the draw doesn't come in often enough.

This principle has many implications for real poker play; the first is that it's frankly terrible to find oneself in a situation with a marked open draw. You can see from this previous example - X only called for value with $^1/_7$ of his hands on the river, instead of $^6/_7$, but Y gained nothing from compensating bluffs. X's strong position on the turn enables him to give up when the flush comes in because Y paid so much for the opportunity to bluff on the river.

In fact, you might notice that Y's partial clairvoyance doesn't help him very much at all here; this occurs primarily because his distribution of hands is so weak. Optimal strategies would normally never result in a situation where a hand was marked as a draw; in fact, our fellow poker theorist Paul R. Pudaite has proposed a "Fundamental Theorem of Chasing" that conjectures that given sufficient potential future action, every action sequence must have some positive chance of containing hands that are the nuts for all cards to come.

The second principle is also of prime importance in understanding how to apply game theory to poker.

Multi-street games are not single-street games chained together; the solution to the full game is often quite different than the solutions to individual streets.

In the first game, we considered the play on the river in isolation as a problem unto itself. However the solution to the two-street game is different, even though after the play on the turn, the second game reduced to an identical situation to the first. In fact, this is almost always the case. This is one of the most important differences in the analysis of games such as chess or backgammon and the analysis of poker. In those games, if we arrive at a particular position, the move order or the history of the game is irrelevant to the future play from this point. This is not the case in poker.

This critically important point also leads us to one of the great difficulties of doing truly optimal poker analysis - no situation exists in a vacuum. There was always action preceding the play on a particular street that should be taken into account. There are always more complex distributions of hands that must be taken into account. In fact, we really cannot label a play or solution "optimal" unless it is taken from the full solution to the game. This difficulty is in some ways insurmountable due to lack of computing power and also because of the multiplayer nature of the game. However, we still look for insight through the solving of toy games and then try to adjust for these factors in real play.

Example 20.3 - The Clairvoyance Game (Mixture of Draws)

Suppose that instead of a distribution of hands for Y that contains only draws and dead hands, he has a distribution which contains x% flush draws (the main draw), y% weak closed draws - for example underpairs that don't have sets yet, and the remainder z% are dead hands. For the sake of simplicity, presume that the flush hits 20% of the time and the weak draws hit 4% of the time (these events are independent - both could happen, or neither).

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The weak draws don't have pot odds (even considering the extra bet they might win on the river) to call a bet on the flop. However, since Y is calling with additional hands in order to balance his bluffs on the river, he might as well (in fact, he must to be optimal) call with all of his weak closed draws first, and finally with some dead hands.

So X bets, and Y calls with a mixture of the three types of hands. However, Y still has some equity on the river when the flush doesn't come in, because 4% of the time he has hit a set with a non-flush card. He will bet his sets for value and bluff with some dead hands. (Busted flush draws, weak draws that didn't come in, and dead hands from the start are all dead hands on the river).

In the previous game (Example 20.2), it was clear which hands X had to make indifferent by calling on the river. X could not prevent Y from profitably calling and betting the river for value when he made his flush. Hence, he made Y indifferent to calling with a pure bluff. Here, however, he must decide between trying to make Y indifferent with his weak closed draws or with his completely dead hands. This choice depends on the total fraction of each of these hands that are present.

If x = 10% and z = 90%, we have the game we previously solved.

Suppose that x = 10% and y = 90% (there are no dead hands). Then clearly X must make Y indifferent with his weak closed draws. He does this by calling enough in both cases that Y is indifferent to calling and bluffing the river. Say that x_f is X's calling frequency when the flush comes in, and x_n is X's calling frequency when another card hits.

Now let's consider a single of Y's closed draws. Y has a number of strategies he can employ. We'll transform multiple decisions into a matrix because it makes solving the game easier. We'll denote Y's strategies as the following:

[<turn action>/<river action if the flush comes in>/< river action if the flush misses >]

So for example [F//] is a turn fold, while [C/B/K] means to call on the turn, bluff the river if the flush comes in and check the river if the flush misses. In all cases, if the hand hits a set, it will value bet on the river. If this occurs, then us equity is 5 bets (the bets in the pot and from the turn) plus the appropriate implied odds (x_f calls if the flush came in, x_n calls if it missed). We'll assume that the set is made with a flush card $^1/_4$ of the time.

$V_{set} = (^3/_4)(5+x_n) + (^1/_4)(5+x_f)$

Y's action sequence	Y's equity	Y's equity assuming $x_n = 1$ and $x_f = 1$
[F//]	$-(^1/_{25})(4) = -0.16$	
[C/B/B]	$(^1/_{25})V_{set} + (^{19}/_{100})(-2x_f+5(1-x_f)) +(^{77}/_{100})(-2x_n+5(1-x_n))$	
[C/B/K]	$(^1/_{25})V_{set} + (^{19}/_{100})(-2x_f+5(1-x_f)) - ^{77}/_{100}$	
[C/K/B]	$(^1/_{25})V_{set} - ^{19}/_{100}+(^{77}/_{100})(-2x_n+5(1-x_n))$	
[C/K/K]	$(^1/_{25})V_{set} - ^{19}/_{100} - ^{77}/_{100}$	$^{72}/_{100}$

The values $^1/_{25}$, $^{19}/_{100}$, and $^{77}/_{100}$ are the respective probabilities of hitting a set (whether a flush comes or not), hitting a flush card but not a set, and missing everything.

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We can immediately see that calling and checking down whatever comes must be dominated. Suppose that xn and xf were both 1, the most favorable situation for a set. Then <[C/K/K]> = $^{1}/_{25}$ (6) - $^{96}/_{100}$ = - $^{72}/_{100}$, which is substantially worse than simply folding immediately.

We also know that the total number of bluffing hands when the flush comes will be greater than the total number of bluffing hands when the flush does not come (because we will have more value hands). Therefore, there must be some hands that are played in the manner [C/B/K] in addition to those which are played in the manner [C/B/B], (If any hands are played [G/K/B]. we can simply remove these hands and convert [C/B/K] to [C/B/B]).

Since we have both types of hands, bluffing on the river when the flush does not come in must he a mixed strategy.

We therefore have:

<[C/B/B]> = <[C/B/K]>
$^{77}/_{100}$ (-2x_n + 5(l - x_n) = - $^{77}/_{100}$
5 - 7x_n = - l

We additionally know that Y must bluff at least sometimes on the river since he does not always have the nuts. Hence we have the following:

$^{4}/_{100}V_{set}$ + $^{19}/_{100}$ (-2x_f + 5(1-x_f)) - $^{77}/_{100}$ = $^{16}/_{100}$
$^{1}/_{25}$ (($^{3}/_{4}$)(5+$^{6}/_{7}$)+($^{1}/_{4}$)(5+x_f))+ $^{19}/_{100}$ (5-7x_f) = $^{61}/_{100}$
x_f = $^{3}/_{7}$

This is a pretty interesting result as well: by adding draws that are only $^{1}/_{4}$ as strong as the flush draws, Y is able to force X to call on the river three times as often. Here we see the power of closed draws. When only closed draws are possible (the flush didn't come in), X must call $^{6}/_{7}$ of the time to keep Y indifferent. When the flush does come in however (an open draw), X needn't call so much, only $^{3}/_{7}$ of the time.

For Y's play, he need only play enough weak draws so that he has the appropriate amount of bluffing hands when the flush comes in. If he plays *y* weak closed draws, he will have a made hand $^{1}/_{9}$ y + $^{1}/_{10}$ of the time on the river, and his a should be $^{1}/_{7}$. Hence,

($^{1}/_{7}$)($^{1}/_{9}$y + $^{1}/_{10}$) = $^{8}/_{9}$y
$^{8}/_{9}$ y = $^{1}/_{63}$ + $^{1}/_{70}$
y = $^{9}/_{550}$

(This value is a little more than $^{1}/_{70}$, the total number of dead hands Y played in the previous game).

Now let's consider a more genera] case, where Y's distribution contains some of each type of hand. The shape of his distribution determines the strategy of the game - X must make a decision regarding which type of hand to make indifferent. Y needs to add "bluffing" hands to support his flush draw calls. Y will of course prefer to add hands with equity compared to hands with none. So he will add the weak closed draws to his calling distribution until they are exhausted. Only then will he add dead hands and attempt to bluff when the open draw comes in.

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In a way, this is similar to the [0,1] game. Y's distribution has regions; a region of flush draws, a region of weak closed draws, and a region of dead hands. He has to take actions in certain frequencies for purposes of balance, to make X unable to exploit him. Likewise, however, he will choose hands that have positive value with which to do this. At some point, there will be a threshold hand that will have a mixed strategy - at that hand he will be indifferent.

Sometimes, that threshold hand will be a weak closed draw (as we saw when there were no dead hands). Sometimes, that threshold hand will be a dead hand (as we saw when there were no weak closed draws). The key is: if there are not enough weak closed draws in Y's distribution, then X cannot make him indifferent to calling and bluffing with those closed draws. Y will always have greater equity from doing that than doing something else. This occurs in just the same way that X cannot make him indifferent to calling with flush draws; X cannot bet enough to prevent him from making money. The presence of the flush draws allows the weak closed draws (up to a certain number) to profitably call on the turn because they obtain positive value (compared to checking) by bluffing when the flush comes in. This positive value enables them to overcome the fact that they are calling without sufficient pot odds to do so.

So suppose that x = 30%, y=1%, and z= 69%. Then X will make Y indifferent to calling and bluffing with a dead hand. Y will simply call with all his weak closed draws. But if x = 30%, y = 50%, and z = 20%, then X makes Y indifferent to calling and bluffing with a weak closed draw.

This is a question that people who are familiar with our work on optimal play often ask us: "Which draws should we make indifferent to calling and bluffing?" The answer is, in a sense, the best one you can! It is possible to extend the above methodology- to include any combination of draws that do not contain the nuts. We can take this concept further by combining the clairvoyance game with the two-street game and looking at a situation where a known, exposed hand plays against a known distribution with cards to come.

Example 20.4

Consider the following situation:

X and Y play holdem on the turn.
X's hand is exposed.
Y holds some hand y chosen randomly from a distribution of hands Y.
There are P units in the pot.

Now we can directly map Y and y to probabilities; Y knows X's hand, and he knows all the possible cards that can come on the river, so his chance of being ahead in a showdown (with no other betting) is simply the probability that he will have the best hand after the river card (for the sake of this discussion, we'll ignore ties).

For example, suppose we have the following:

A's hand: A♠ 4♣
The board: A♥ J♠ 8♦ 7♣
Y's distribution of hands: (99+, AQ+}
Y has the following types of hands:

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Hand Type	Frequency	Probability of winning
AA	$^{1}/_{39}$	1
KK	$^{6}/_{39}$	$^{2}/_{44}$
QQ	$^{3}/_{39}$	$^{2}/_{44}$
JJ	$^{3}/_{39}$	1
TT	$^{6}/_{39}$	$^{6}/_{44}$
99	$^{6}/_{39}$	$^{6}/_{44}$
AK	$^{8}/_{39}$	$^{41}/_{44}$
AQ	$^{6}/_{39}$	$^{41}/_{44}$
Total	1	

We can sum the frequencies of each of the unique probability outcomes (without regard to the actual hand type) and create an alternate distribution $\overline{\boldsymbol{Y}}$, which for the purposes of this game is equivalent to $\boldsymbol{Y}$:

$\{(1, {}^{4}/_{39}), ({}^{41}/_{44}, {}^{14}/_{39}), ({}^{6}/_{44}, {}^{12}/_{39}), ({}^{2}/_{44}, {}^{9}/_{39})\}$

The reason that this is equivalent to the previous distribution is that all the hands with one probability of winning are interchangeable, even though they have different cards that "make" them, and even though both players will see the river card. This is because on the river, the second player has perfect information about the hand. Therefore, he will never pay off a bet incorrectly, so the first player cannot gain from inferences about the river card by betting or raising.

There are, of course, two streets remaining, but play on the river is fairly simple. If the hand gets to the river, the player with the made hand checks (if he is first). The player with the distribution of hands then value bets all his winning hands, plus a fraction of losing hands equal to his winning hands times $1/(P + 1)$. This is the familiar betting-to-bluffing ratio that we saw first in the one-street clairvoyance game (Example 11.1). Then the opponent calls $P/(P+ 1)$ of the time with his exposed hand, and there is a showdown.

Y therefore breaks even on his bluffs, and wins one bet when he is called with a winner. The bets he wins on the river are "implied odds" from the play on the turn, but because the play on the river is essentially automatic if both players play optimally, we can actually incorporate that directly into calculations about play on the turn. We define the value of a winning concealed hand to be the pot size on the river plus $P/(P + 1)$ bets, the amount that the concealed hand wins by value betting and bluffing appropriately on the river. We use the notation P^t to represent this value.

$P^t = P+P/(P+1)$

X has to pay off here with $P/(P+ 1)$ of his hands because we constrained the game to contain only a half street on the turn.

Let's assume that in the above example after the turn betting there are five bets in the pot. Now X will bet his winning hands, and be called ${}^{5}/_{6}$ of the time. So the value of the hand is the pot (5 bets) plus ${}^{5}/_{6}$ of a bet = ${}^{35}/_{6}$.

Now using our mapped distribution, we can define some more things. Suppose we order the probabilities from best to worst. So in the example distribution, the probabilities are:

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$^{4}/_{39}$ of the hands have equity 1.
$^{14}/_{39}$ of the hands have equity $^{41}/_{44}$
$^{12}/_{39}$ of the hands have equity $^{6}/_{44}$
$^{9}/_{39}$ of the hands are have equity $^{2}/_{44}$

We now create a function that identifies the probability of a particular hand based on the rank of that hand:

y(t) = the winning probability of the best hand in $\overline{\boldsymbol{Y}}$*, after removing the t best hands from* $\overline{\boldsymbol{Y}}$*.*
(0 <= t < = 1)

y(0) = the winning probability of the best hand in $\overline{\boldsymbol{Y}}$.
y(1) **=** the winning probability of the worst hand in **Y'**.

In our example distribution, we have:

y(0) = 1
y($^{1}/_{4}$) = $^{41}/_{44}$
y($^{1}/_{2}$) = $^{6}/_{44}$ (the median hand)
y(1) = $^{2}/_{44}$

Another function that is quite useful for analyzing this game is the average equity of the best t hands, which we denote by $y^{\theta}(t)$. This is a measure of the probability of a range of hands; it is the expected value of the designated subset of the distribution. So for example, $y^{\theta}(^{1}/_{2})$ is the average of *y(t)* for the top 50% of hands.

$y^{\theta}(t)$ (the value of the tth hand) is often discontinuous and ill-behaved.

We can see some values of the $y^{\theta}(t)$ function for our example distribution:

$y^{\theta}(0) = 1$
$y^{\theta}(^{1}/_{4}) = (^{16}/_{156})(1)+(^{23}/_{156})(^{41}/_{44}) \approx 0.23995 / 0.25 \approx 0.95980$
$y^{\theta}(^{1}/_{2}) = (^{8}/_{78})(1) + (^{28}/_{78})(^{41}/_{44})+(^{3}/_{78})(^{6}/_{44}) \approx 0.44231/.5 \approx 0.88462$
$y^{\theta}(1) \approx 0.48951$

y ¯(t) is a much different and much more well-behaved function; it turns out that this function has some nice properties.

- Continuous.
- Decreasing for all *t*, and strictly decreasing as long as y(t) <> y(0).

Another way of expressing $y^{\theta}(t)$ (which shows its continuity) is as follows:

$$y^{\theta}(t) = \left(\int_0^t y(t)dt\right)/t$$

$y^{\theta}(1)$ is an important value - the average equity for all the hands is **Y**. This value will be quite significant in determining the play. For example, suppose that $y^{\theta}(1) = 1$. Then Y would bet the turn and X would fold (because X has no equity in the pot!). Similarly, if
$y^{\theta}(1) = 0$, then Y could not win and X would simply claim the pot. However, between these two extremes, there are cutoffs such that the players employ mixed strategies designed to prevent exploitation.

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Example 20.5 - The One-and-a-Half-Street Clairvoyance Game

One half-street on the turn (X checks dark), followed by a full street on the river.
Y's hand is exposed.
Y's hand is randomly chosen from a distribution Y
There are P bets in the pot.
No raising is allowed*

* There is one extremely limited case where Y might bet and X check-raise if it were allowed, when the relationship between the distribution of Y, the adjusted pot on the turn, and the adjusted pot on the river meets a quite narrow set of criteria. We ignore this case for simplicity's sake here by simply removing the raise option.

To solve this game, we must identify the optimal strategies. Consider intuitively how the play should go. If Y's distribution is very strong (such that Y doesn't have "enough" equity), then will just bet and take the pot, as X lacks pot odds to call. On the other hand, if Y's distribution contains a mixture of strong and weak hands, Y will bet some set of his better hands and check the rest. And if Y's distribution contains only weak hands, then he will just check and play the river.

First, let's find some of the threshold values. We'll calculate all equities from Y's point of view as usual. Assume Y bets all hands. If he is called, the pot on the river will be

$P+2$ bets and his hand will be worth $(P+2)^t$ on the river if he hits a whining card. It costs him one bet to make this play. The betting on the river is already included in the ^{t}. If Y bets and X folds, then Y wins P units.

X should fold immediately if the value of folding is less than the value of calling.

$y^\theta(1)(P+2)^t - 1 \geq P$
$y^\theta(1) \geq (P+1)/(P+2)^t$
or
$y^\theta(1) \geq (P+1)/(P+3)/(P+2)(P+4)$

So if Y's overall hand strength (chance of winning) from all the hands in his distribution is greater than the value $(P+1)/(P+2)^t$, then Y can bet and claim the pot. X can't profitably call on the turn at all.

We'll call that Case 1.

Case 1.
Condition: $y^\theta(1) \geq (P+1)/(P+2)^t$
Action: Y bets and X folds.

Something else comes up here. Suppose Y's distribution is such that this case is satisfied, but checks and then bets the river? Can X sometimes call then?

The answer is no. The reason is that by doing so, X can be exploited. Suppose that Y has $^1/_5$ hands that always win, and the remaining $^2/_5$ of hands are close to always winning, such that Case l's condition is true. If Y can occasionally check the nuts on the turn, bet the river and occasionally get called, then he can employ the following exploitive strategy:

Bet all hands that don't have 100% equity (X must fold) and win P units.

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Check the nuts and bet the river.
Hope to get called and win more than *P* units.

Since X's optimal strategy cannot allow this exploitation. X's optimal strategy is to fold at any future point during the hand. This is actually a general rule:

If your opponent's distribution and yours are such that your opponent can take some action and you must fold and surrender the entire pot, then at any later point in the hand, it is at least co-optimal to fold to any future action.

Avoiding this situation is another reason to play properly balance strategies throughout the hand.

In any event, at the opposite end of the spectrum, let us consider the situation where X should always call a bet. If X will always call, then Y's equity from checking a handy is simply $y(P^t)$. That is, Y checks the turn and if he wins on the river, the value of the hand is P^t. Y's equity from betting a hand *y* is similarly $y(P+2)^t - 1$. Then Y should bet when:

$y(P+2)^t - l \geq y(P^t)$
$y \geq 1/((P+2)^t - P^t)$

What this means is that if X will call all the time, then Y can profitably bet all hands such that *y* is greater than this value. For most P values, this expression is a little less than $^1/_2$. At P= 3, for example, it evaluates to around 0.48. It should be clear that Y profits by betting when he has more than 50% equity in the pot. However, from this expression, we can see that Y can profitably bet with a little less than 50% equity hi the pot, because of the implied odds his clairvoyance affords him on the next street.

Now we can make use of one more piece of notation. We will often want to designate "the set of hands where *y* is greater than some value y_0," in terms of how many hands there are or what fraction of the hands satisfy this condition. So the set of hands $\{y \mid y \geq y_0\}$ corresponds to a different set $\{y(t) \mid t \leq \mathcal{T}\}$ for some value $\mathcal{T}$. We'll call this value $\mathcal{T}[y_0]$. Let's go back to our example distribution **Y'**:

$\{(1, {}^4/_{39}), ({}^{41}/_{44}, {}^{14}/_{39}), ({}^6/_{44}, {}^{12}/_{39}), ({}^2/_{44}, {}^9/_{39})\}$

Suppose we want all the hands where x is greater than $y_0 = {}^1/_2$. This set of hands includes all the 100% winners and all the ${}^{41}/_{44}$ winners {AA, JJ, AK. AQ}, We can also define this set of hands with a t value that corresponds to the cutoff value, t values below this threshold will have equity greater than or equal to y_0, and t values above this threshold will have less. Since these hands make up ${}^{18}/_{39}$), of the total, $\mathcal{T}[{}^1/_2] = {}^{18}/_{39}$. This is the *t* value such that $y(t) \geq {}^1/_2$ for all *t* values less than $\mathcal{T}[{}^1/_2]$, or more straightforwardly, it's the fraction of hands with equity greater than or equal to $^1/_2$.

$y({}^1/_2)$ is the equity of the best hand left after removing the best $^1/_2$ of hands. So there are 39 hands total. Removing the best 19 of them, the best hand left is ${}^6/_{44}$. So that's what y(t) is, the best hand after removing the best t hands. $\mathcal{T}[x]$ is like the inverse of this. So suppose we have an *equity* $^1/_2$, and we want to know how many hands have at least that much equity. In this case it's ${}^{18}/_{39}$. $\mathcal{T}[x]$ is the fraction of hands with at least x equity.

So in this case, we'll define:

$t_0 = \mathcal{T}[(1/((P+2)^t - P^t)]$

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t_0 is the threshold where if X calls all the time. Y should bet all hands that have greater equity. If Y does bet all these hands, then X can call all the time as long as Y's equity with the hands he bets is less than $(P+1)/(P+2)^t$.

To summarize this case, if X calls all the time, then Y can come up with a set of hands to bet with a threshold of t_0. His equity with that set of hands is $y^\theta(t_0)$. As long as this value is low enough that X always has odds to call on the turn, X just calls all the time.

Case 2.
Condition: $y^\theta(t_0) \leq (P+1)/(P+2)^t$
$t_0 = \mathcal{T}[(1/((P+2)^t - P^t)]$
Action: Y bets and B calls.

So those are the two extreme cases - where Y has almost all of the equity, and where Y doesn't have enough strong hands to prevent X from calling no matter what strategy he employs.

The third case, however, is the most interesting one, because X utilizes a mixed calling strategy. That is, the set of hands that Y bets is too strong for X to call all the time, but not strong enough for X to simply fold. Instead, X calls often enough that Y can't just bluff his bad hands with impunity, but folds often enough that Y can't exploit him by betting strong hands exclusively.

So X is going to have some "folding frequency," which is the % of the time he will fold on the turn. We will call this value α. Here, however, since it is the turn, applying the typical $\alpha = P/(P+1)$ is incorrect. Remember, X's hands are all alike, so his play can be expressed as a single number (call/fold frequency on the turn). On the river, of course, he simply calls $P/(P + 1)$ of the time. Y's strategy, on the other hand, includes betting different types of hands with different values.

If X will fold α of the time and call 1- α of the time, we can find Y's strategy in terms of α. Y will bet when the equity of betting is higher than the equity of checking.

$\langle Y, \text{bet hand } x \rangle = \alpha P + (1 - \alpha)\,(y\,(P + 2)^t - 1)$
$\langle Y, \text{check hand } x \rangle = yP^t$
$\alpha P + (1 - \alpha)\,(y\,(P + 2)^t - 1) \geq yP^t$
$y\,[(1 - \alpha)(P+ 2)^t - P^t] \geq 1 - \alpha(P+ 1)$

The left side of this equation is positive as long as $\alpha P < 2$, or $\alpha < 2/P$.

The right side of this equation is positive as long as $\alpha > 1/(P+ 1)$. So if X chooses his folding frequency a greater than $1/(P + 1)$, then Y just bets all the time. This could be optimal (in Case 1 from above), but if that case doesn't apply, then Y is exploiting X. We can also see that if X chooses a frequency a that's high enough, then Y bets whenever his equity from betting is higher than checking:

$y\,[(1 - \alpha)(P+ 2)^t - P^t] \geq 1 - \alpha\,(P+1)$

$$y \geq (1 - \alpha\,(P+1))/[(1 - \alpha)(P+2)^t - P^t] \qquad \textbf{(20.1)}$$

We can use another fact about optimal strategies. If X is playing a mixed strategy of calling and folding with his hand, then he must be indifferent to doing those two things. Otherwise, he could simply improve by moving his calling frequency. So suppose that Y is betting his optimal set of hands $\mathcal{T}$. Then we have!

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$y^{\theta}(\mathcal{T})(P+2)^t - 1 = P$
$y^{\theta}(\mathcal{T}) = (P+1)/(P+2)^t$

Now we return to the properties of $y^{\theta}(t)$. This function is decreasing, so we know that $y^{\theta}(1) \leq y^{\theta}(t) \leq y^{\theta}(0)$. We also have an important value $(P + 1)/(P + 2)^t$. This value is the value of $y^{-}(t)$ such that if Y bets all hands better than T, X is indifferent to calling or folding. So one of the following must be true.

Possibility 1: $y^{\theta}(1) \geq (P+ 1)/(P+ 2)^t$
Possibility 2: $y^{\theta}(0) \leq (P \setminus 1)/(P+ 2)^t$
Possibility 3: $y^{\theta}(\mathcal{T}) = (P+ l) /(P+ 2)^t$

Possibility 1 is within Case 1, where Y bets all the time and X just folds.

Possibility 2 is within Case 2, where Y only bets his hands that have positive equity, and X calls all the time.

Possibility 3 is a new case, Case 3. We know that $y(t) \leq y^{\theta}(t)$, because y(t) is just the worst hand that makes up the equities in $y^{\theta}(t)$. If $y(\mathcal{T}) > l/((P + 2)^t - P^t)$: then the threshold hand (the hand that makes X indifferent to calling) is still a "good hand." So Y bets and X still calls because he still has odds, and this is actually Case 2.

If $y(\mathcal{T}) < l/((P+2)^t - P^t)$, however, then we have Equation 20.1, which we found earlier, and shows that Y will only bet hands that have positive equity from betting:

$y \geq (1 - \alpha(P + 1))/ [(1 - \alpha) (P + 2)^t - P^t]$

Substituting $y(\mathcal{T})$ for y and solving for α, we get:

$\alpha = (1 - y(\mathcal{T})[(P+ 2)^t - P^t])/(P + 1 - y(t)(P + 2)^t)$

In this case, Y chooses a $\mathcal{T}$ such that X is indifferent to calling or folding against the whole range of hands better than T. And X calls and folds with a frequency that makes Y indifferent to betting with exactly $y(\mathcal{T})$.

Case 3.
Condition: Neither Case 1 nor Case 2 hold.
Y bets hands such that $y^{\theta}(\mathcal{T}) = (P+ 1)/(P+ 2)^t$
X folds $\alpha = (1 - y(\mathcal{T})[(P + 2)^t - P^t])/(P + 1 - y(t)(P + 2)^t)$ of his hands and calls with the rest.

Now let's look at some examples. First, we return to our example hand. Recall that the situation was:

X's hand: A♠ 4♣
The board: A♥J♠ 8♦ 7♣
Y's distribution of hands: {99+, AQ+}

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Y has the following types of hands:

Hand Type	**Frequency**	**Probability of winning**
AA	$^{1}/_{39}$	1
KK	$^{6}/_{39}$	$^{2}/_{44}$
QQ	$^{3}/_{39}$	$^{2}/_{44}$
JJ	$^{3}/_{39}$	1
TT	$^{6}/_{39}$	$^{6}/_{44}$
99	$^{6}/_{39}$	$^{6}/_{44}$
AK	$^{8}/_{39}$	$^{41}/_{44}$
AQ	$^{6}/_{39}$	$^{41}/_{44}$
Total	1	

This converted into a secondary Y distribution of:

$\{(1, {}^{4}/_{39}), ({}^{41}/_{44}, {}^{14}/_{39}), ({}^{6}/_{44}, {}^{12}/_{39}), ({}^{2}/_{44}, {}^{9}/_{39})\}$

Say the pot is 5 bets.
First of all, we can look for which case we are in.

Case 1:
$y^{\theta}(1)$ is about 0.48 (as we found earlier)
$(P+1)/(P+2)^t = (6)/({}^{63}/_{8}) = {}^{48}/_{63}$, or about ${}^{3}/_{4}$

So Case 1 doesn't hold because $y^{\theta}(1) \geq (P+1)/(P+2)^t$.

This should make sense, because Y's distribution here isn't strong enough that he can just blast away with all his hands and X has to fold. X has a big share of the pot if Y bets all hands, so he won't fold outright.

Case 2:
First, we can find t_0.

$t_0 = \mathcal{T}\ [(1/((P+2)^t) - P^t)] = \mathcal{T}[{}^{8}/_{15}]$

$\mathcal{T}[{}^{8}/_{15}]$ is the t value for which $y(t)$ is greater than ${}^{8}/_{15}$. That includes all the hands with 100% equity and all the hands with ${}^{41}/_{44}$ equity, or ${}^{18}/_{39}$ of the hands. If the following is true, we have Case 2:

$y^{\theta}(t0) \leq (P+1)/(P+2)^t$
$y^{\theta}({}^{18}/_{39}) \leq {}^{48}/_{63}$

The average equity of the top ${}^{18}/_{39}$ of Y's hands is just:

$y^{\theta}({}^{18}/_{39}) = ({}^{4}/_{18})(l) + ({}^{14}/_{18})({}^{41}/_{44}) \approx 0.94697$

Case 2 isn't true either, because the inequality above isn't satisfied. This also should make sense. Y has some pretty strong hands in his distribution in addition to a lot of weaker ones. If X were to call all the time, Y could selectively bet his strong hands to exploit X.

Since neither Case 1 nor Case 2 hold, we must be in Case 3.

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Now what we must do is find a $\mathcal{T}$ value for Y, such that if Y bets, X is indifferent to calling or folding.

$y^{\theta}(\mathcal{T}) = (P+ l) /(P+ 2)^{t}$
$y^{\theta}(\mathcal{T}) = {}^{48}/_{63}$

We can work backwards into this:

$\{(1, {}^{4}/_{39}), ({}^{41}/_{44}, {}^{14}/_{39}), ({}^{6}/_{44}, {}^{12}/_{39}), ({}^{2}/_{44}, {}^{9}/_{39})\}$

${}^{48}/_{63} = [({}^{4}/_{39})(1) + ({}^{14}/_{39})({}^{41}/_{44}) + (y)({}^{6}/_{44})]/({}^{18}/_{39}+ y)$

(where y is the number of ${}^{6}/_{44}$ equity hands that Y will bet)

${}^{48}/_{63}({}^{18}/_{39}+ y) = [({}^{4}/_{39})(1) + ({}^{14}/_{39})({}^{41}/_{44}) + (y)({}^{6}/_{44})]$

$y \approx 0.13655$
$T = y + {}^{18}/_{39} \approx 0.59808$

(You may wish to independently verify that $y^{\theta}(\mathcal{T}) = {}^{48}/_{63}$)

So Y bets his best 59.8% of hands (approximately). We must also find X's calling frequency. (He seeks to make Y indifferent between bluffing and checking at T).

$\alpha = (1 - y(\mathcal{T})[(P+2)^{t} - P^{t}])/(P + 1 - y(t)(P+2)^{t})$
$\alpha = (1 - ({}^{6}/_{44})({}^{15}/_{64}))/(6 - ({}^{6}/_{44})({}^{63}/_{8}))$
$\alpha \approx 0.19651$

So X folds about 19.6% of his hands.

This game has three important features that help to inform our decisions in real poker.

- Implied odds are an important part of optimal strategy in games with more than one street. These games are often more difficult to solve because of their increased complexity. Hence, game theory examples often consider the play on a single street and show how to make the opponent indifferent on the one street. In this game, the P^{t} idea introduces a quantitative measure of the implied odds. We can see from this that the strategy for each individual street is not the same as the strategy for that street in isolation. For example, on the turn, the pot contains 5 bets. Usually, if a player folds more often than $1/(P+ 1)$ of the time, he is exploitable by bluffing. However, in this case, with a pot size of 5, it is correct for X to fold almost ${}^{1}/_{5}$of his hands (instead of ${}^{1}/_{6}$). This is because X is at a disadvantage not only on this, but on future sneets. When he folds, he doesn't have to pay off on the river. Hence, he can fold more often optimally. We cannot emphasize this concept enough - multi-street games are not single-street games chained together; the solution to the full game is often quite different than the solutions to individual streets.
- The distributions that the players hold cause the play to be significantly different. When Y is quite strong, X can just fold; when Y is weaker, X can just call all the time. X can sometimes play accurately against Y's distribution.
- The players cannot make each other indifferent to taking certain actions with certain hands, hi fact, there is usually just one threshold where they are indifferent, but that

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threshold forms the linchpin of the entire strategy. In this case, the values for T and α represent those thresholds. X can vary the value of α fairly widely without giving up value as long as Y continues to play optimally. However, if X strays from the optimal α value, Y can exploit him by bluffing or not as appropriate.

More Advanced Topics In Games With Draws

Tire previous game was a fairly general case of a game where X could not bet on the turn. We have done some work on games with even more complexity; we omit describing some of these games because they are substantially more difficult, and often involve many cases related to the shapes of the distributions. However, we will briefly touch upon some of the principles we see from studying more complex games.

One important principle we find throughout looking at games with different types of draws is that that concentrating equity in a few hands is stronger than having a distribution that contains many draws of moderate value. Contrast the following two distributions - one that contains 25% the nuts and 75% dead hands and one that contains 100% hands that have 25% equity. In terms of showdown equity, these distributions are identical; *y(1)* for both of them is 0.25. However, the distribution that contains nut hands can play much more effectively in a two-street game, extracting additional value from his winning hands. When he holds a homogeneous set of draws, he basically has to wait until the draw comes in or doesn't in order to effectively bet and bluff. Distributions that contain more hands with high showdown value are in general more like clairvoyance than distributions that contain hands with medium-strength. And clairvoyance is effectively the limit of how well you can do playing a game.

As a result, it is important, as we said earlier, to keep one's distributions balanced with different types of hands, and to allow nut hands to balance draws whenever possible when deciding to carry hands from street to street.

In full two-street games we also see semi-bluff raising. When Y's distribution is sufficiently weak or draw-heavy, X will bet out on the turn. Y should then raise with strong hands and balance those raises with semi-bluffs, while continuing to call with weaker draws and folding the weakest draws of all (beyond the bluffing hands he needs to carry forward, as in the first games we studied in this chapter),

A third principle we see is that in clairvoyant situations, open draws are often much weaker than closed draws. When a draw is closed, the non-clairvoyant player must fold just α of his hands. Contrast this to the results we saw in open draw games, where the non-clairvoyant player was able to fold more than half his hands, even when there were some closed draws present.

The complexity of multi-street games provides a tremendous challenge for us as poker analysts; there are many factors to be considered about the shape of the distributions, the size of the pot. and the structure of the game. And these are only two-street games where the bet size is the same on each street. If we had a full understanding of all the ways in which distributions made up of hands of varying strengths and varying draw potential interacted across multiple streets in optimal play, we would have a solution to any poker game. As it is, we simply continue to try to take out tiny chips from that large tree, by analyzing one situation at a time.

Key Concepts

- There are two principles at the core of analyzing multi-street play. The first is that in order for a bluff to be credible, it must come from a distribution of hands that contains strong hands as well. The second principle is also of prime importance in understanding how to apply game theory to poker. It is that multi-street games are not single-street

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games chained together; the solution to the full game is often quite different than the solutions to individual streets.

- If a player has a distribution that contains only draws and does not threaten to have a strong hand in a particular situation, it is typically very bad for his expectation.
- Closed draws are much more powerful than open draws because the opponent is forced to pay off to value bets more often.
- Some draws (or made hands) are so strong that it is impossible to make them indifferent to calling a value bet. When choosing which draw to make indifferent, the answer is often the strongest draw that you can.
- Implied odds have a very important impact on action frequencies when draws are possible. In the clairvoyant game with arbitrary draws, the entire game revolves around equity in the implied pot - which contains additional value once the river is completed.
- If your opponent's distribution and yours are such that your opponent can take some action and you must fold and surrender the entire pot, then at any later point in the hand, it is at least co-optimal to fold to any future action.

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